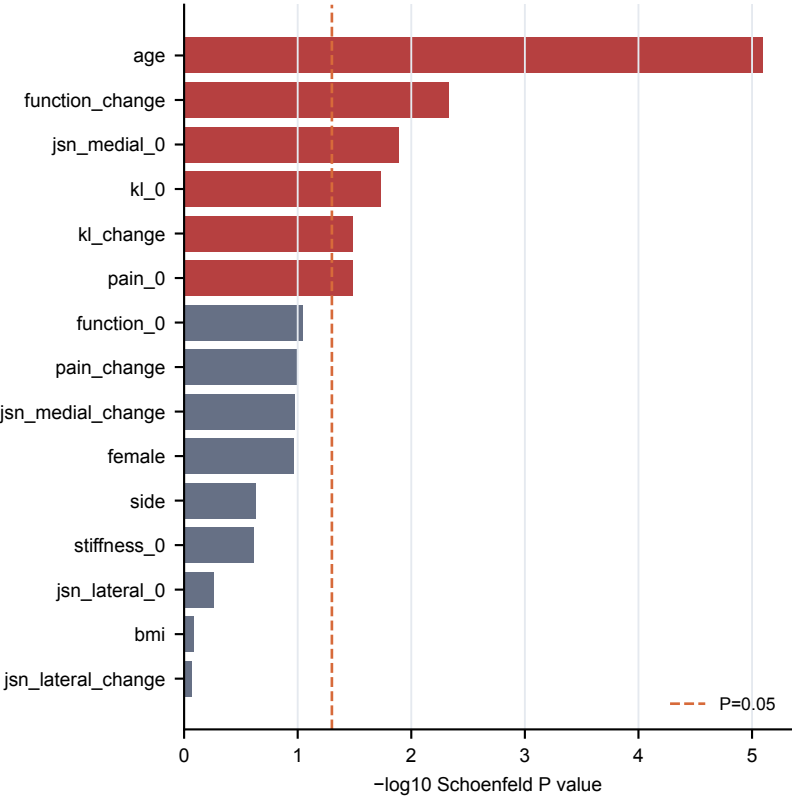


Supplementary Figure S7 | Model E proportional-hazards sensitivity

a Predictor-level PH diagnostics



b Apparent fixed-horizon sensitivity

Metric	Average-effect	Time-varying
C-index	0.770	0.770
60-mo AUC	0.797	0.801
60-mo Brier	0.109	0.108
Calibration intercept	-0.049	-0.033
Calibration slope	1.132	1.019

Time-varying terms: age, baseline pain, function change, baseline KL, KL change, and baseline medial JSN.
Interaction: $x \times \log(\max[t, 0.1])$.